

Curve fitting using numpy and scipy

```
In [9]: import numpy as np
        from scipy.optimize import curve_fit
        import matplotlib.pyplot as plt
```

```
In [10]: # given
x = [1,2,3,4,5,6]
y = [7,25,30,45,45,60]
```

```
In [11]: def f(x, a, b):
        z = a*x + b
        return z
```

```
In [12]: par, err = curve_fit(f, x, y)
a = par[0]
b = par[1]
print("slope is: ", a, "y-intercept is: ", b)
```

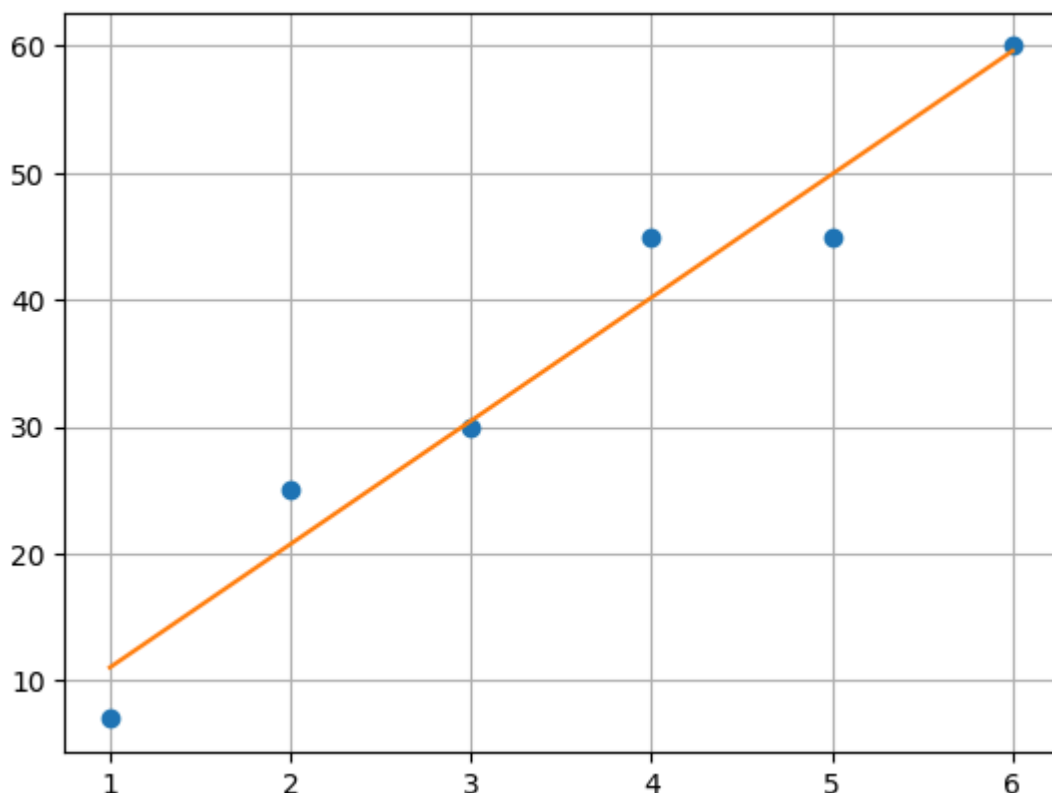
slope is: 9.714285714285712 y-intercept is: 1.33333333333333428

```
In [13]: x_new = np.linspace(1, 6, 100)
y_new = f(x_new, a, b)

plt.plot(x, y, "o")

plt.plot(x_new, y_new)

plt.grid()
plt.show()
```



Curve fitting using only numpy package

```
In [14]: x1 = np.array(x)
y1 = np.array(y)

m,c = np.polyfit(x, y, 1)

print("slope is: ", m, "y-intercept is: ", c)
```

slope is: 9.714285714285714 y-intercept is: 1.3333333333333337

```
In [15]: x1_new = np.linspace(1, 6, 100)
y1_new = m*x1_new + c

# print(x1_new, y1_new)
```

```
In [16]: plt.plot(x1, y1, "o")

plt.plot(x1_new, y1_new)

plt.grid()
plt.show()
```

